

Linear_regression_analytical

September 14, 2025

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[1]: import numpy as np
```

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[2]: # =====  
# Example dataset (replace with your data)  
# =====  
# X: features (rows = examples, columns = features)  
# y: target values  
X = np.array([[1, 2],  
              [2, 3],  
              [4, 5],  
              [6, 7]], dtype=float)  
  
y = np.array([3, 5, 9, 13], dtype=float)  
  
# =====  
# 1. Add bias column (theta_0)  
# =====  
X_bias = np.hstack([np.ones((X.shape[0], 1)), X]) # shape: (m, n+1)
```

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[3]: # =====  
# 2. Compute theta using normal equation  
# =====  
XTX = X_bias.T @ X_bias  
  
# Check if XTX is invertible  
if np.linalg.matrix_rank(XTX) == XTX.shape[0]:  
    theta = np.linalg.inv(XTX) @ X_bias.T @ y  
else:  
    # Use pseudo-inverse if singular  
    theta = np.linalg.pinv(X_bias) @ y  
  
print("Theta (coefficients including bias):", theta)
```

Theta (coefficients including bias): [4.4408921e-16 1.0000000e+00 1.0000000e+00]

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[4]: # =====  
# 3. Make predictions  
# =====
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y_pred = X_bias @ theta  
print("Predicted y:", y_pred)
```

Predicted y: [3. 5. 9. 13.]